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5 point summary

Date types(variable type)

The basic distinction is between

quantitative variables (for which one asks “how much?”) and

categorical variables (for which one asks “what type?”).

(1) Quantitative variables

(a)Discrete number- counted as whole no.

exp:- No. of kids 2, 3, Training session 1,2,3 its always integer not a float.(2.5)

(b)Continuous:- Number can be infinite precision .

Exp:-you can sleep 7 hr 30 min & 20 sec, distance can be 70.97 meters

(2) Categorical variables

Can not be measure against each other like color (blue, red, yellow)

(4)Ordinal

-Result rank A++,A+,A,B+.. here it can be compare among them.

Examples of types of data	
Quantitative	
Continuous	Discrete
Blood pressure, height, weight, age	Number of children Number of attacks of asthma per week
Categorical	
Ordinal (Ordered categories)	Nominal (Unordered categories)
Grade of breast cancer Better, same, worse Disagree, neutral, agree	Sex (male/female) Alive or dead Blood group O, A, B, AB

Variables shown at the left of Table can be converted to ones further to the right by using “cut off points”.

For example, blood pressure can be turned into a nominal variable by defining “hypertension” as a diastolic blood pressure greater than 90 mmHg, and “normo-tension” as blood pressure less than or equal to 90 mmHg.

Height (continuous) can be converted into “short”, average” or “tall” (ordinal).

nominal

Some examples of variables that can be measured on a nominal scale include:

- **Gender:** Male, female
- **Eye color:** Blue, green, brown
- **Hair color:** Blonde, black, brown, grey, other
- **Blood type:** O-, O+, A-, A+, B-, B+, AB-, AB+
- **Political Preference:** Republican, Democrat, Independent
- **Place you live:** City, suburbs, rural

Variables that can be measured on a nominal scale have the following properties:

- **They have no natural order.** For example, we can't arrange eye colors in order of worst to best or lowest to highest.
- **Categories are mutually exclusive.** For example, an individual can't have *both* blue and brown eyes. Similarly, an individual can't live *both* in the city and in a rural area.
- **The only number we can calculate for these variables are *counts*.** For example, we can count how many individuals have blonde hair, how many have black hair, how many have brown hair, etc.
- **The only measure of central tendency can be use is *the mode*.** The mode tells us which category had the most counts.

Ordinal

The next type of measurement scale that we can use to label variables is an **ordinal scale**.

Ordinal scale: A scale used to label variables that have a natural *order*, but no quantifiable difference between values.

Some examples of variables that can be measured on an ordinal scale include:

- **Satisfaction:** Very unsatisfied, unsatisfied, neutral, satisfied, very satisfied
- **Socioeconomic status:** Low income, medium income, high income
- **Workplace status:** Entry Analyst, Analyst I, Analyst II, Lead Analyst
- **Degree of pain:** Small amount of pain, medium amount of pain, high amount of pain

Basic measurement of Statistics

Mean, median, Mode, Variance, Standard Deviation, Quartile Range...

df.describe():-function to summarize information

```
In [55]: df1.describe()
```

Out[55]:

	Profit	Unit Price	Shipping Cost
count	17.000000	18.000000	17.000000
mean	305.741382	75.768333	17.985294
std	884.219143	119.998788	23.028008
min	-481.041000	2.880000	0.700000
25%	-11.682000	4.987500	2.990000
50%	26.920000	39.415000	5.260000
75%	313.578000	99.722500	26.220000
max	3424.220000	500.980000	74.350000

- **Counts :-No. of observations**
- **Average value (mean)**
- **Std:-On average, how much each measurement deviates from the mean (standard deviation of the mean)**
- **Min:- minimum value of the observation.**
- **Max:- Maximum value of the observation.**
- **25,50,75 percentiles. (Quartile Range)**
- **Most frequently occurring value (mode)**
- **Midpoint between the lowest and highest value of the set (median)**

Mean :-

Mean is the average of the given data set.

Mean for ungrouped data = Sum of observations/Number of observations

1. Find the mean of the first 10 odd integers.

Solution:

First 10 odd integers: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19

Mean = Sum of the first 10 odd integers/Number of such integers

$$= (1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19)/10$$

$$= 100/10$$

$$= 10$$

Therefore, the mean of the first 10 odd integers is 10.

Median:-

The median is the middlemost data value of an ordered data set. This can be estimated using the formulas given below.

When n is odd:

$$\text{Median} = (n + 1)/2\text{th observation}$$

When n is even:

$$\text{Median} = [(n/2)\text{th observation} + \{(n/2)+1\}\text{th observation}]/2$$

2. What is the median of the following data set?

32, 6, 21, 10, 8, 11, 12, 36, 17, 16, 15, 18, 40, 24, 21, 23, 24, 24, 29, 16,
32, 31, 10, 30, 35, 32, 18, 39, 12, 20

Solution:

The ascending order of the given data set is:

6, 8, 10, 10, 11, 12, 12, 15, 16, 16, 17, 18, 18, 20, 21, 21, 23, 24, 24, 24, 29, 30, 31, 32, 32, 32, 35, 36, 39, 40

Number of values in the data set = $n = 30$

$$n/2 = 30/2 = 15$$

15th data value = 21

$$(n/2) + 1 = 16$$

16th data value = 21

Median = $[(n/2)\text{th observation} + \{(n/2)+1\}\text{th observation}]/2$

$$= (15\text{th data value} + 16\text{th data value})/2$$

$$= (21 + 21)/2$$

$$= 21$$

Mode:-

Mode is the most frequently occurring value in the data set.

3. Identify the mode for the following data set:

21, 19, 62, 21, 66, 28, 66, 48, 79, 59, 28, 62, 63, 63, 48, 66, 59, 66, 94, 79, 19, 94

Solution:

Let us write the given data set in ascending order as follows:

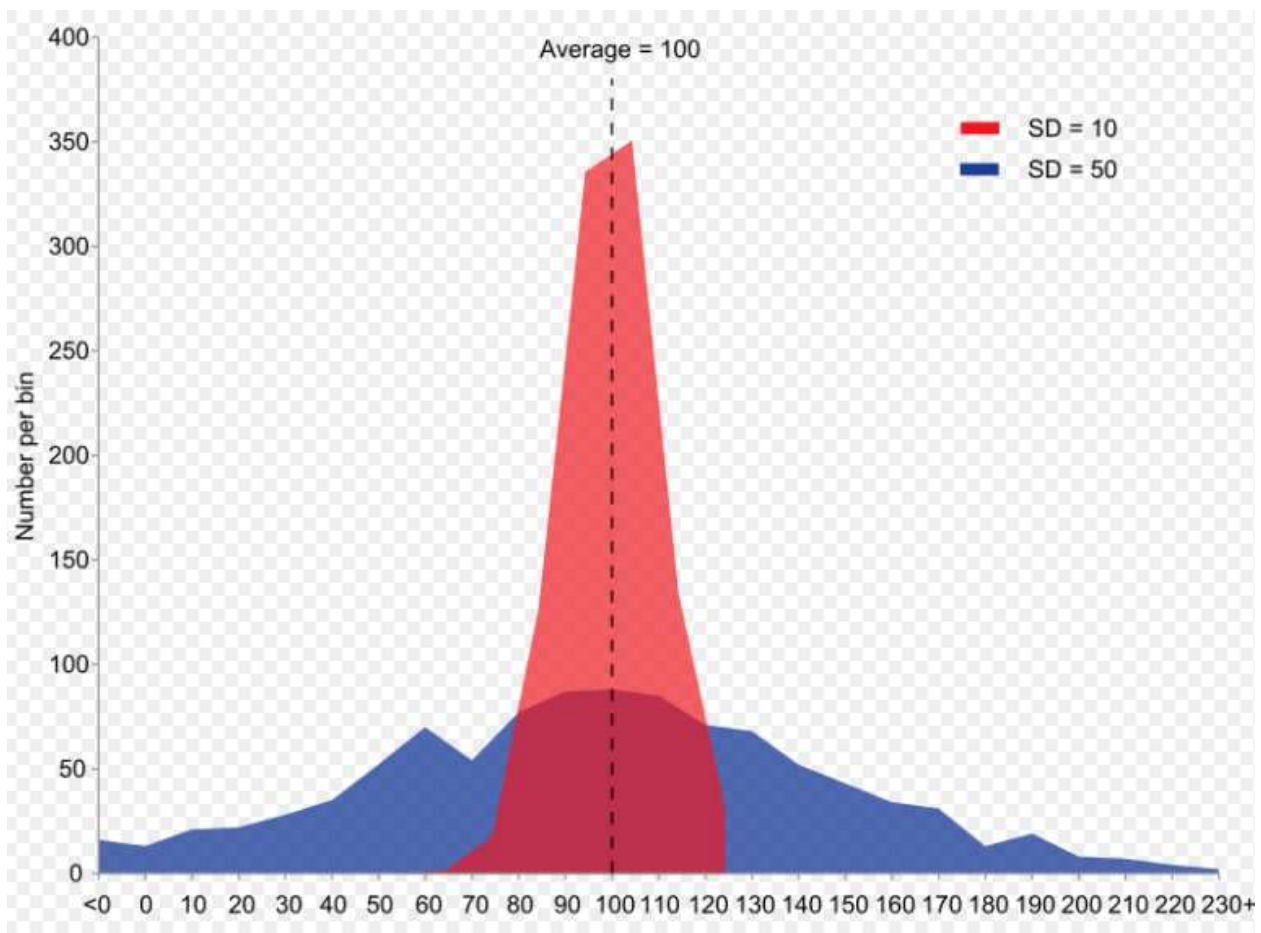
19, 19, 21, 21, 28, 28, 48, 48, 59, 59, 62, 62, 63, 63, 66, 66, 66, 66, 79, 79, 94, 94

Here, we can observe that the number 66 occurred the maximum number of times. Thus, the mode of the given data set is 66.

Variance

Dispersion

Dispersion (also called variability, scatter, or spread) is the extent to which a distribution is stretched or squeezed. Common examples of measures of statistical dispersion are the variance, standard deviation, and interquartile range. For instance, when the **variance** of data in a set is **large**, the data is **widely scattered**. On the other hand, when the **variance** is **small**, the data is less scattered.



Example:-

Cricketer IPL Match

Component life of an equipment

Pizza Delivery

Variance of Population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

σ^2 = Population Variance
 N = the size of the population
 x_i = each value from the population
 μ = the population mean

Standard Deviation: -

It is the square root of variance.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation
 N = the size of the population
 x_i = each value from the population
 μ = the population mean

standard deviation is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean (also called the expected value) of the set, while a high standard deviation indicates that the values are spread out over a wider range.

If the deviation is large, it means the numbers are spread out, further from the mean or average.

Example Problem to calculate variance and Standard Deviation.

In a chemical experiment You grow 20 crystals from a solution and measure the length of each crystal in millimeters. Here is the data:

9, 2, 5, 4, 12, 7, 8, 11, 9, 3, 7, 4, 12, 5, 4, 10, 9, 6, 9, 4

Calculate the population standard deviation of the length of the crystals.

Approach: -

1. Calculate the mean of the data.

$$(9+2+5+4+12+7+8+11+9+3+7+4+12+5+4+10+9+6+9+4) / 20 = 140/20 = 7$$

2. Subtract the mean from each data point and squaring this number, (so it does not matter if it is positive or negative).

$$(9 - 7)^2 = (2)^2 = 4$$

$$(2 - 7)^2 = (-5)^2 = 25$$

$$(5 - 7)^2 = (-2)^2 = 4$$

$$(4 - 7)^2 = (-3)^2 = 9$$

$$(12 - 7)^2 = (5)^2 = 25$$

$$(7 - 7)^2 = (0)^2 = 0$$

$$(8 - 7)^2 = (1)^2 = 1$$

$$(11 - 7)^2 = (4)^2 = 16$$

$$(9 - 7)^2 = (2)^2 = 4$$

$$(3 - 7)^2 = (-4)^2 = 16$$

$$(7 - 7)^2 = (0)^2 = 0$$

$$(4 - 7)^2 = (-3)^2 = 9$$

$$(12 - 7)^2 = (5)^2 = 25$$

$$(5 - 7)^2 = (-2)^2 = 4$$

$$(4 - 7)^2 = (-3)^2 = 9$$

$$(10 - 7)^2 = (3)^2 = 9$$

$$(9 - 7)^2 = (2)^2 = 4$$

$$(6 - 7)^2 = (-1)^2 = 1$$

$$(9 - 7)^2 = (2)^2 = 4$$

$$(4 - 7)^2 = (-3)^2 = 9$$

3. Calculate the mean of the squared differences.

$$(4+25+4+9+25+0+1+16+4+16+0+9+25+4+9+9+4+1+4+9) / 20 = 178/20 = 8.9$$

This value is the variance. The variance is 8.9

4. The population standard deviation is the square root of the variance.

$$(8.9)^{1/2} = 2.983$$

The population standard deviation is 2.983

Sample Variance vs. Population Variance: What's the Difference?

The formula to calculate **population variance** is:

$$\sigma^2 = \sum (x_i - \mu)^2 / N$$

where:

- Σ : A symbol that means “sum”
- μ : Population mean
- x_i : The i^{th} element from the population
- N : Population size

The formula to calculate **sample variance** is:

$$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$$

where:

- $\bar{x}$: Sample mean
- x_i : The i^{th} element from the sample
- n : Sample size

	Mean	Standard Deviation	Variance	Size
Population	μ	σ	σ^2	N
Sample	$\bar{x}$	s	s^2	n

When we calculate population variance, we divide by N (the population size).

When we calculate sample variance, we divide by $n-1$ (the sample size – 1).

When calculating the sample variance, we apply something known as Bessel's correction – which is the act of dividing by $n-1$.

dividing by $n-1$ corrects the bias in the estimation of the population variance.

Here $(n-1)$ is the **Degrees of freedom: - refers to the maximum number of values, which have the freedom to vary, in the data sample.**

Q. The following sample shows 10 different values for the voltage of a cable:

57, 64, 68, 66, 54, 60, 48, 46, 75, 62

Find the mean and the variance.

Ans:-

$$\begin{aligned}\text{Ans} \quad \bar{X} &= \sum_{i=1}^n \frac{X_i}{n} = \\ &= \frac{57 + 64 + 68 + 66 + 54 + 60 + 48 + 46 + 75 + 62}{10} \\ &= \frac{600}{10} = 60 \\ S^2 &= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n \cdot \bar{X}^2 \right) \\ \sum_{i=1}^n X_i^2 &= 57^2 + 64^2 + 68^2 + 66^2 + 54^2 + 60^2 + 48^2 \\ &\quad + 46^2 + 75^2 + 62^2 = 36\,730 \\ n \cdot \bar{X}^2 &= 10 \times 60^2 = 36\,000 \\ S^2 &= \frac{1}{9} (36\,730 - 36\,000) = \frac{730}{9}\end{aligned}$$

Same answer will get if apply this formula also: -

$$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$$

When to Calculate Sample Variance vs. Population Variance

Rule of thumb: -

- calculate the **sample variance** when the dataset we're working with represents a sample taken from a larger population of interest.
- calculate the **population variance** when the dataset we're working with represents an entire population, i.e. every value that we're interested in.

Example: Calculating Sample Variance

Suppose a botanist wants to calculate the variance in height of a certain species of plants. Because there are thousands of individual plants in one region, she decides to take a simple random sample of 20 plants and measure each of their heights.

In this scenario, the botanist should calculate the **sample variance** because she is interested in the variance of the entire population of plants.

Example: Calculating Population Variance

Suppose a teacher wants to calculate the variance of exam scores for the 20 students in her class.

In this scenario, the teacher should calculate the **population variance** because the dataset she's working with (the 20 exam scores) represent the entire population that she is interested in.

5 point summary

Quantile

(MINIMUM)

Q1- first quantile

Q2-Median (2nd quantile)

Q3- third quantile

(MAXIMUM)

After arranging the dataset in ascending order, when we find the position of the median, the position of median is dividing the dataset in to two group.

Q1= Median position of the first half is first quantile.

Q3= Median position of the second half is Third quantile.

Q2= Median of the complete dataset .

Inter Quantile Range

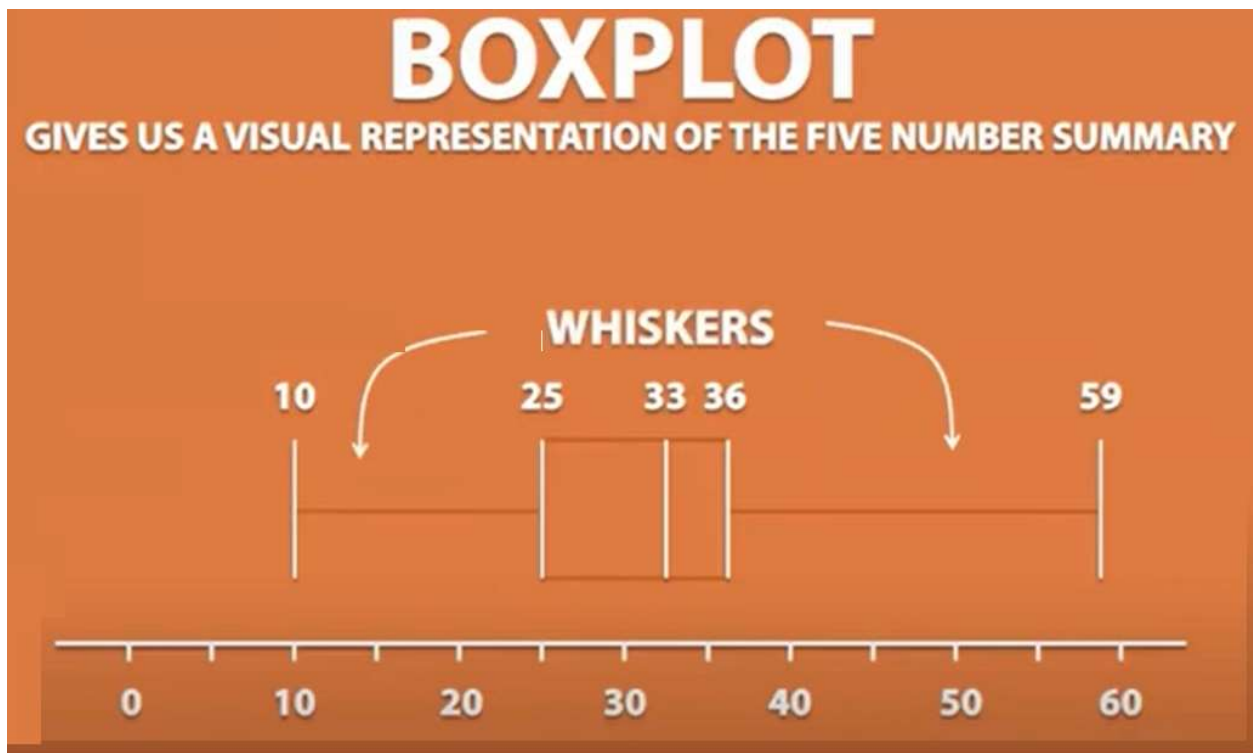
$IQR = Q3 - Q1$.

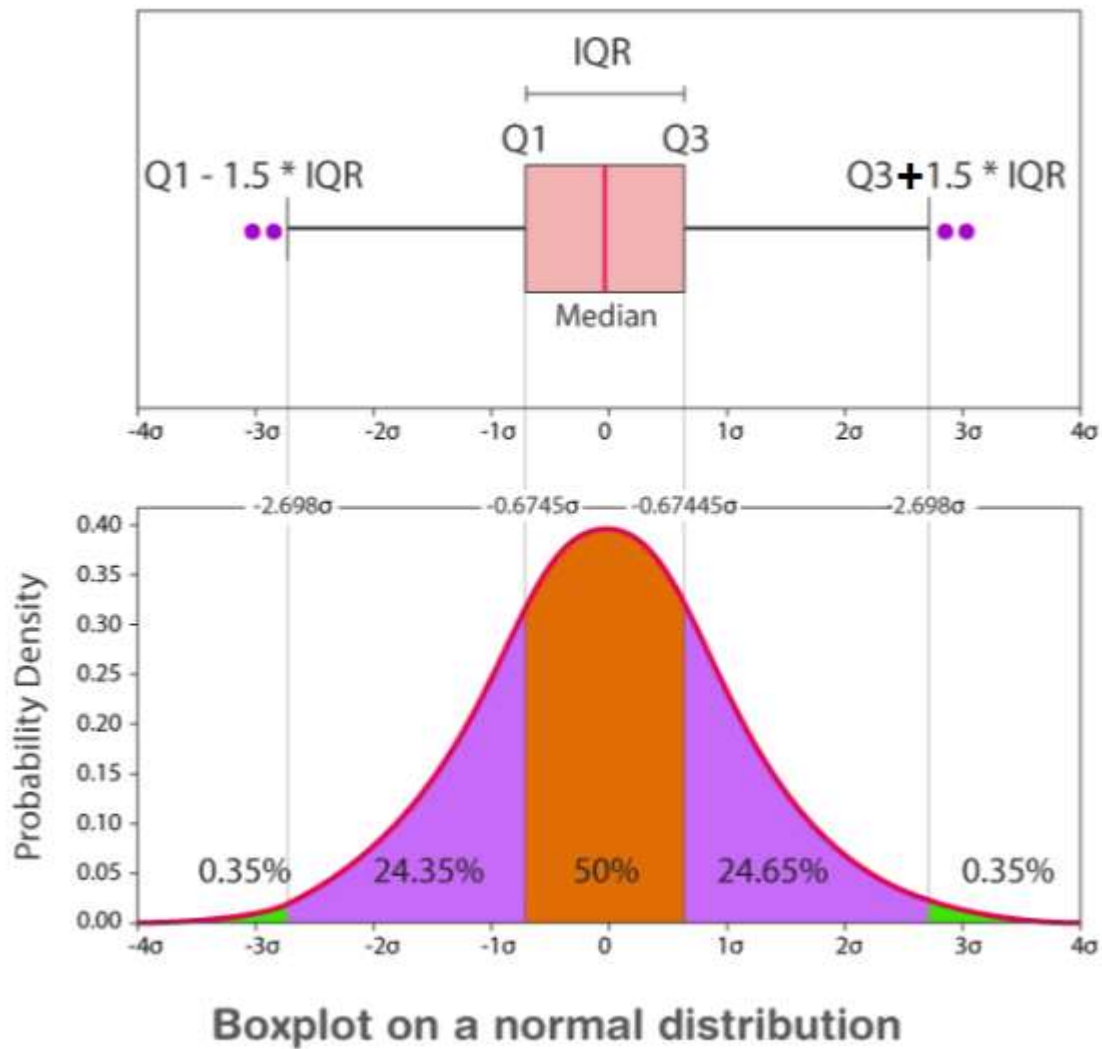
Find Q1, Q2, Q3, IQR?

12 25 25 27 10 11 31 33 36 43 50 59 34 34 35

Data point arranged in ascending order,

10 11 12 25 25 27 31 33 34 34 35 36 43 50 59





Also called Gaussian Distribution

Similar to bell shape curve....

Normal Distribution



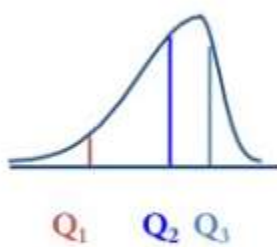
Positive Skew



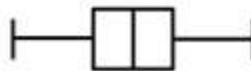
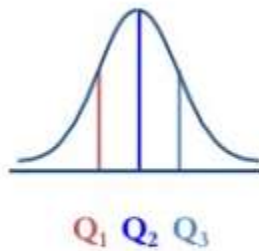
Negative Skew



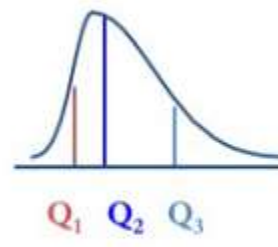
Left-Skewed



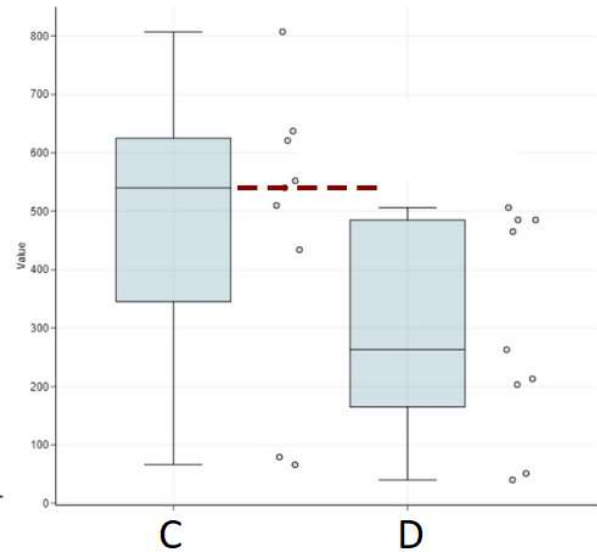
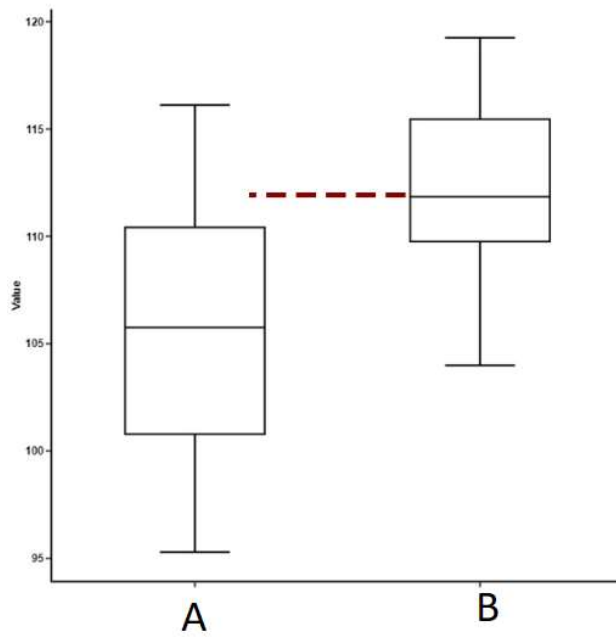
Symmetric



Right-Skewed



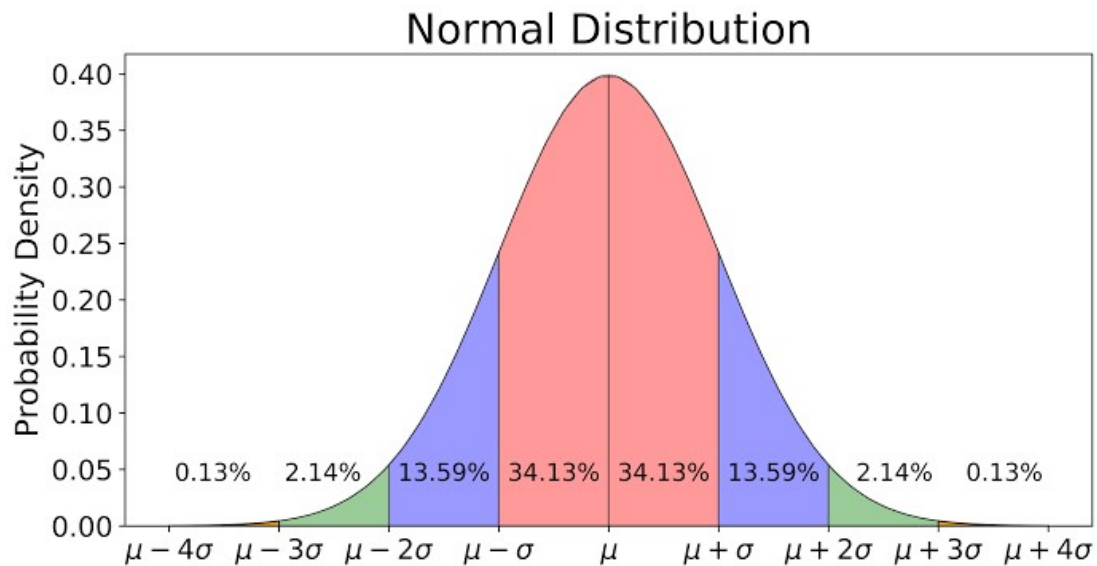
Compare between different variable BOX PLOTS



Empirical Rule.

BELL shape and symmetric.

A useful continuous distribution is the standard normal distribution with mean equal to 0 and standard deviation equal to 1



Normal distribution is a density curve total area equals to 100%.

There are certain observations which could be inferred from this Rule 68,95,99. Called **Empirical Rule**.

About 68.26% data lies within 1 SD ($<\sigma$) of the mean (μ),
About 95.44% data lies within 2 s d ($<\sigma$) of the mean (μ),
About 99.72% data lies within 3 s d ($<\sigma$) of the mean (μ)
& the rest 0.28% of the data lies outside 3 SD ($>3\sigma$) of the mean (μ), And this part of the data is considered as outliers.